

Hot Hand, Real or Believed?

A behavioural analysis of setting decisions in professional volleyball — Scientific Essay

Studies on Human Behaviour — Course Project

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GitHub repository: https://github.com/FraMichi/Studies_on_human_behaviour.git

Abstract. The hot hand is the belief that a successful action makes another success more likely. While this phenomenon has been widely studied, most evidence on how this belief affects allocation decisions comes from controlled experiments. This study investigates the hot hand in a real volleyball setting, using 63 Data Volley scouting files from one season of two Italian Serie A2 women's teams. A total of 8,312 attacks are analysed using logistic mixed-effects models with random intercepts for player and match.

The results show no evidence of a real hot-hand effect for either team once differences between players are taken into account. The analysis of the setter's decisions also shows that rotation is an important factor: controlling for it removes an apparent avoidance pattern in one team and reveals a significant tendency to select a successful attacker again in the other. Finally, there is no evidence of worse attacking performance under score pressure. High-ball use is strongly related to reception quality, suggesting that fewer high balls in critical phases should not be interpreted simply as a more risky attacking strategy.

Overall, the results show the importance of considering both player behaviour and the specific structure of volleyball when analyzing the hot hand using observational data.

1. Introduction

Anyone who has played, watched, or coached a team sport has probably heard the same sentence: “she’s on fire tonight.” The idea behind it is simple: a player who has just succeeded may be more likely to succeed again. This belief, usually known as the hot hand, has been widely studied in behavioural science, but the evidence is more complex than it may initially seem.

The hot hand can actually be considered from two different perspectives. The first concerns performance: does a player's probability of success really increase after a previous success? The second concerns behaviour: do the people making decisions around that player act as if this effect exists? These two questions do not necessarily have the same answer. A decision-maker may believe in the hot hand even when there is little evidence of an actual change in performance.

Volleyball provides an interesting setting to study this distinction because the setter repeatedly decides which teammate will receive the ball for the next attack. These choices, together with the outcomes of each action, are recorded by the scouting software used by volleyball teams. This makes it possible to compare actual attacking performance with the setter's allocation decisions using data from real matches.

2. The Hot Hand in the literature

2.1 Early research on the hot hand

The modern debate on the hot hand started with Gilovich, Vallone, and Tversky (1985). They analysed shooting sequences from professional and college basketball and found no evidence that a successful shot increased the probability of success on the next one. Despite this, players and fans strongly believed that shooting streaks existed.

The authors explained this belief as a misconception of randomness: people tend to expect even short random sequences to contain a balanced mix of outcomes, and therefore perceive clusters of successes as meaningful streaks. The hot hand was consequently interpreted as a cognitive illusion, and this view remained influential for many years.

2.2 The hot hand in volleyball

One important criticism of the original basketball studies concerns the role of the defence. When a basketball player starts scoring repeatedly, the opposing team can adapt its strategy, for example by increasing defensive pressure on that player. A real improvement in performance could therefore be partly hidden by the reaction of the defence.

Raab, Gula, and Gigerenzer (2012) studied the hot hand in volleyball, where the interaction between attacker and defense is different. Their analysis of first-division players found evidence of performance streaks for some players. They also found that coaches could identify changes in players' performance and that playmakers used this information when deciding whom to set. The authors therefore interpreted hot-hand decisions as potentially adaptive rather than simply irrational.

2.3 Hot-hand beliefs and decision-making

Other studies have investigated more directly whether the hot-hand belief affects allocation decisions.

Köppen and Raab (2012) used controlled video sequences and found that participants tended to allocate more balls to a player presented as being on a hot streak and fewer to a player on a cold streak.

MacMahon, Köppen, and Raab (2014) further showed that hot-hand beliefs can depend on factors such as framing and previous experience with the sport. This suggests that the belief is not necessarily fixed, but can change depending on how performance information is presented and interpreted.

More recently, Ittlinger, Lang, Schubert, and Raab (2025) studied judgement in elite beach volleyball and found that players' estimates of their chances of winning could be affected by cognitive biases. Although this study does not directly examine the hot hand, it provides further evidence that decisions and judgments in volleyball can be influenced by the competitive context.

2.4 Research gap

Previous research therefore provides mixed evidence about whether the hot hand represents a real performance effect, while experimental studies show that the belief can influence allocation decisions. However, studying these decisions in real matches introduces additional factors that are easier to control in experiments. In volleyball, for example, the setter cannot always choose among the same attackers because player availability changes with rotation.

This study examines the relationship between previous success and real setting decisions using observational match data. In particular, it considers whether an apparent hot-hand pattern remains after accounting for structural factors such as rotation.

3. Research Questions

The project separates three questions that are often considered together in discussions of the hot hand:

RQ1 — Real hot hand. Does an attacker's probability of succeeding depend on whether her immediately preceding attack in the same match was successful?

RQ2 — Believed hot hand. Is a successful attacker more likely to be selected again on the team’s next attack, after accounting for differences between players, matches, and rotation?

RQ3 — Competitive pressure. Does the critical phase of a set affect attack-error probability or the use of high balls?

The two setters under study (Vittoria Prandi for Brescia and Francesca Scola for Talmassons) played in the same league and season, allowing their allocation patterns to be compared in similar competitive contexts.

4. Data and Method

4.1 Dataset

The dataset consists of 63 Data Volley (.dvw) scouting files from the 2025/2026 season of two Italian Serie A2 women’s teams, Valsabbina Millenium Brescia and CDA Volley Talmassons FVG, both promoted to Serie A1 at the end of the season.

The dataset includes 36 Brescia files and 27 Talmassons files. Since the two head-to-head matches are stored only once, under Brescia, Talmassons appears in 29 matches in total. The matches cover the regular season, Coppa Italia, and the promotion play-offs.

Parsing produced 93,884 scouted actions, of which 48,699 were performed by the two focal teams.

4.2 Parsing and derived variables

Files were parsed using pydatavolley (OpenVolley project), an open-source Python library that returns one row for each scouted action, including team, player, skill, evaluation, score, set number, attack-combination code, and player positions.

Attack tempo is not directly available in the standard output, so it was decoded from the [3ATTACKCOMBINATION] section of each .dvw file. Attacks were classified as quick (Q), super (U), medium (M), high (H), or other (O). Tempo was successfully identified for all attacks.

As an example, a single attack entry in the raw file is shown below:

```
*16AU#VS~45~H2;s;r;;;;20.26.39;1;2;2;1;2074;;4;11;16;3;18;5;9;16;12;1;18;3;
```

The string before the first semicolon contains the action code, which is decoded using the pydatavolley library and the information contained in the DataVolley attack-combination table. In this example, * identifies the home team, 16 is the player’s jersey number, A indicates the skill (attack), and # indicates a point-scoring attack. The attack combination code VS is matched to the corresponding entry in the [3ATTACKCOMBINATION] table. In this case, VS corresponds to zone 4, a right-side approach, and Super tempo (U).

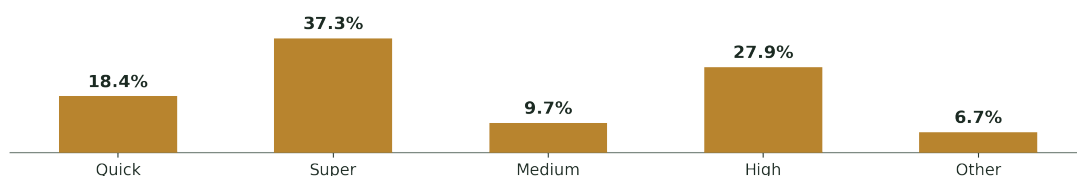


Figure 1. Distribution of attack tempo across the season, own-team attacks ($n = 8,312$), after decoding the attack-combination codes.

Success was defined as a direct point (#) and error as the error code (=). After cleaning, 8,312 attacks by 23 players formed the analytical base.

Field	Code	Meaning
zone	2	Zone 2 (opposite hitter / position 2)
zone	3	Zone 3 (middle hitter)
zone	4	Zone 4 (outside hitter / position 4)
zone	8	Zone 8 (pipe attack)
zone	9	Zone 9 (back-row attack from the diagonal)
tempo	Q	Quick attack
tempo	H	High ball
tempo	U	Super / medium tempo
tempo	M	Medium tempo
tempo	O	Other
tempo	N	No attack / rare case
tempo	T	Fast tempo
side	R	Right-side approach
side	L	Left-side approach
side	C	Middle approach

Table 1. DataVolley code definitions for attack zone, tempo, and approach side.

For RQ1, each attack was linked to the outcome of the same player's previous attack in the same match, giving 7,861 usable observations. For RQ2, each attack was linked to whether the same player was selected again on the team's next attack, giving 8,247 observations. For RQ3, an attack was classified as critical when at least one team had reached 20 points in sets 1–4, while the entire fifth set was considered critical.

RQ2 also considers rotation, since the attackers available to the setter can change after a rally. Rotation was identified by comparing the own team's setter position at the current and next team attack.

Observations with the required rotation information were available for 8,182 attacks.

4.3 Statistical approach

Observations from the same player and the same match are not independent. Some players have a higher baseline probability of success, and matches may also differ in difficulty. A simple comparison of raw proportions could therefore partly reflect these differences rather than the effect of interest.

Each research question was analyzed separately for the two teams using a logistic mixed-effects model with crossed random intercepts for player and match:

$$\text{logit } P(\text{outcome}) = \beta_0 + \beta_1(\text{predictor}) + u_{\{\text{player}\}} + u_{\{\text{match}\}}$$

The two random effects are crossed rather than nested because the same player appears in multiple matches, while each match includes several players. This allows the effect of the predictor to be estimated while accounting for baseline differences between players and matches.

The models were estimated in Python using an iterative procedure implemented with NumPy. Before applying it to the volleyball data, the implementation was tested on simulated data with known parameters.

5. Results

5.1 RQ1 — performance after previous success

A comparison of raw proportions shows a higher success rate after a previous success for both teams: 40.2% vs. 36.8% for Brescia and 42.1% vs. 40.4% for Talmassons.

However, after accounting for differences between players and matches, neither effect is statistically significant (Brescia: $\beta = +0.085$, $SE = 0.064$; Talmassons: $\beta = +0.031$, $SE = 0.071$).

The player-level variance is not negligible, particularly for Brescia ($\sigma^2 \approx 0.11$ on the log-odds scale), confirming that attackers differ in their baseline probability of scoring. Overall, the mixed-effects models provide no evidence of a real hot-hand effect in either team.

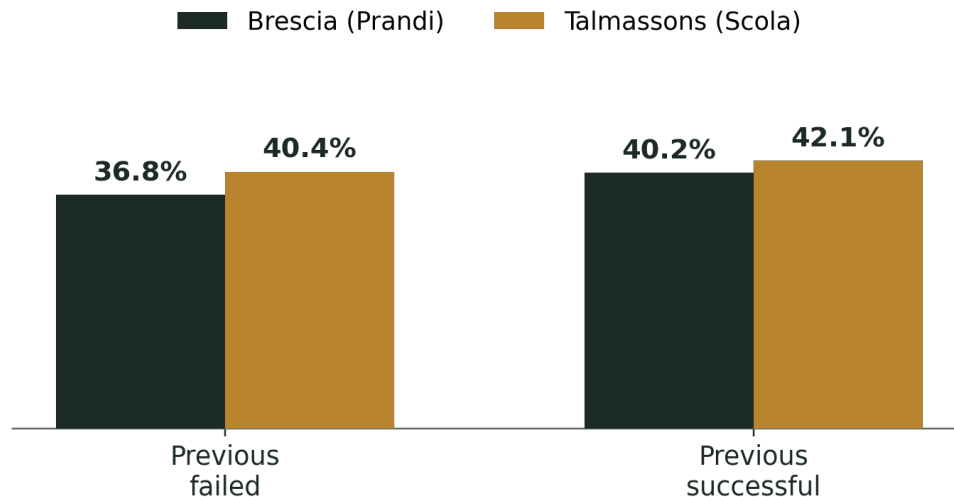


Figure 2. Attack success rate by outcome of the same hitter's previous attack (raw rates). The difference observed for Brescia is not statistically significant in the mixed-effects model.

The difference observed in the raw proportions may therefore be partly explained by player heterogeneity. Players with higher baseline success rates are naturally more likely to appear in the “previous attack succeeded” condition, which can create an apparent pattern when observations are aggregated across players.

This result differs from Raab et al. (2012), who found evidence of streaks for about half of the elite volleyball players in their study. However, there are important differences between the two analyses. Their sample focused on top first-division players, while the present study considers two Serie A2 teams. More importantly, Raab et al. defined streaks using sequences of two or three consecutive successes or failures, while this analysis considers only the immediately previous attack.

A hot-hand effect that develops over longer sequences may therefore not be captured by the present approach. The conclusion should be limited to the definition used here: in this dataset, there is no statistically significant relationship between the outcome of one attack and the next attack by the same player.

5.2 RQ2 — initial allocation results

The initial results suggest different allocation patterns between the two teams. For Talmassons, the same attacker is selected again after a success in 21.4% of cases, compared with 27.3% after a non-success. This difference remains statistically significant after accounting for player and match ($\beta = -0.295$, $SE = 0.082$, $p < .01$).

For Brescia, the probability of selecting the same attacker again is almost identical after a success and a non-success (31.8% vs. 31.7%; $\beta = +0.047$, $SE = 0.067$).

At this stage, the results would suggest an allocation pattern for Talmassons in the opposite direction to a traditional hot-hand belief, while no clear pattern appears for Brescia. However, this interpretation does not yet account for rotation.

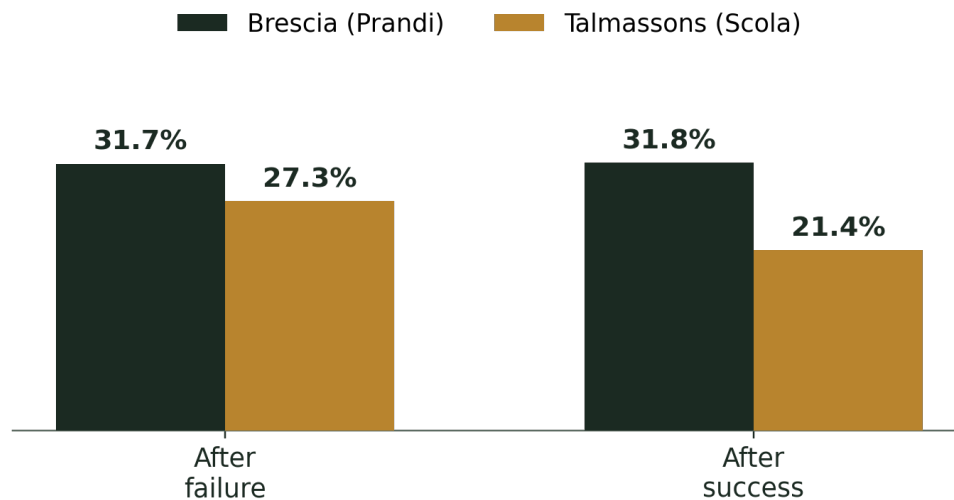


Figure 3. Rate at which the same hitter is selected again on the team's next attack, according to the outcome of her previous attack (raw rates).

5.3 RQ2 revisited — controlling for rotation

This initial result changes when rotation is considered. A team rotates when it wins a rally while receiving, so a successful attack is often followed by a rotation. This happens after about 70% of successful attacks (70.2% for Brescia and 69.3% for Talmassons), compared with about 15% after unsuccessful attacks (15.0% and 14.5%, respectively).

Rotation is also associated with a lower probability of selecting the same attacker again. The re-targeting rate decreases from 34.0% to 27.4% for Brescia and from 27.4% to 20.8% for Talmassons when a rotation occurs. Previous success and the attacking options available on the next action are therefore partly linked by the rules of the game.

After adding rotation to the model, the RQ2 results change. For Talmassons, the negative effect becomes smaller and is no longer statistically significant ($\beta = -0.283$, $p < .01$ without rotation on the same sample; $\beta = -0.142$, $SE = 0.098$ with rotation). For Brescia, the effect becomes positive and statistically significant ($\beta = +0.043$, not significant without rotation; $\beta = +0.246$, $SE = 0.081$, $p < .01$ with rotation).

Rotation itself has a negative coefficient for both teams (Brescia: $\beta = -0.380$, $SE = 0.083$; Talmassons: $\beta = -0.263$, $SE = 0.100$), confirming that it is related to a lower probability of selecting the same attacker again.

The revised RQ2 results therefore show no significant hot-hand allocation pattern for Talmassons. For Brescia, however, a successful attacker is more likely to be selected again after accounting for player, match, and rotation, which is consistent with a hot-hand allocation pattern.

5.4 Interpreting high-ball use

The interpretation of high-ball use in RQ3 requires some caution. A high ball could initially be considered a more conservative attacking option, but in volleyball its use strongly depends on the quality of the previous action. Faster attacks such as quick and super tempos generally require a good reception or defensive action, while a poor first contact often limits the setter's available options and makes a high ball more likely.

This relationship can also be observed directly in the data. When each side-out attack is linked to the reception that preceded it in the same rally, high-ball use changes strongly with reception quality. After a perfect reception, high balls are almost never used, while after a poor reception they are used in around 60% of attacks (Figure 4).

High-ball use should therefore not be interpreted simply as a deliberate choice to play more cautiously. It also reflects the quality of the build-up leading to the attack.

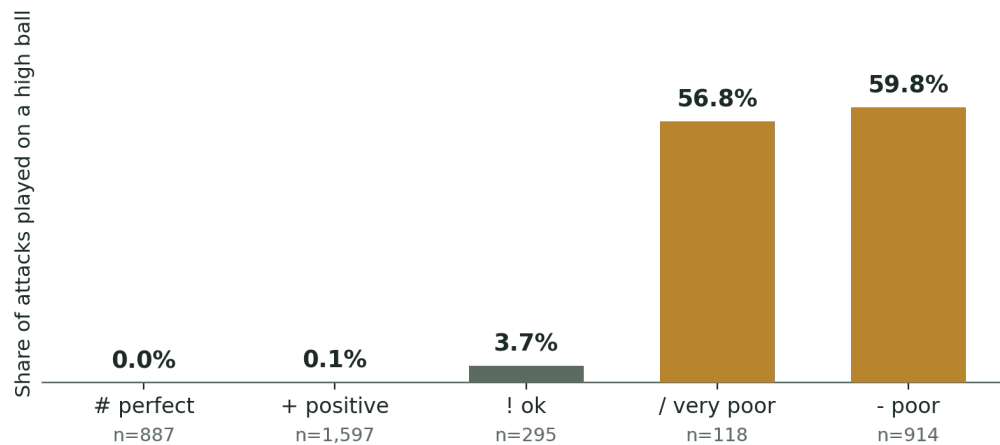


Figure 4. Share of attacks played on a high ball according to the Data Volley evaluation of the preceding reception. The figure includes side-out attacks only (3,811 of 8,312 own-team attacks); the remaining attacks occur in transition after a dig or freeball.

The RQ3b coefficient should therefore not be interpreted as a measure of attacking caution. Instead, high-ball use can be considered an indirect indicator of the quality of the build-up before the attack. A lower probability of high-ball use during critical phases may therefore reflect better conditions for faster attacking options, rather than a greater willingness to take risks.

5.5 RQ3 — effects of score pressure

Overall, 27.3% of own-team attacks (2,273 of 8,312) occur during a critical phase. Talmassons shows no statistically significant change in either attack-error probability ($\beta = -0.046$, $SE = 0.144$) or high-ball use ($\beta = -0.112$, $SE = 0.094$). Brescia, instead, shows significantly fewer attack errors ($\beta = -0.358$, $SE = 0.145$, $p < .05$) and a lower probability of using a high ball ($\beta = -0.190$, $SE = 0.078$, $p < .05$) during critical phases. Considering the relationship between high-ball use and the quality of the preceding actions, the second result should not be interpreted as Brescia becoming more willing to take risks. Instead, the lower use of high balls

is consistent with the team reaching the attack in better conditions during critical phases. Together with the lower error probability, there is no evidence of worse attacking performance under score pressure. If this interpretation is correct, reception quality could also be expected to improve during critical phases. Descriptively, positive receptions increase from 56.2% to 57.9%, but this difference is not statistically significant ($\beta = +0.077$, $SE = 0.093$). Moreover, high-ball use also depends on defensive actions in transition. The proposed explanation is therefore consistent with the results, but it cannot be established from the current models.

One additional point concerns the high-ball model. Player-level variance is much larger than in the other models because high-ball use strongly depends on playing role. For example, middle blockers rarely attack high balls. The player random intercept accounts for these baseline differences, so the coefficient mainly compares each player with herself across normal and critical phases, rather than simply reflecting differences in high-ball use between roles.

Table 1 summarises the mixed-effects model results across all research questions.

Model	Brescia (Prandi)	Talmassons (Scola)	Reading
RQ1 — success ~ prior success	+0.085 (SE 0.064)	+0.031 (SE 0.071)	No significant hot hand effect
RQ2 — same attacker selected ~ success	+0.043 (SE 0.068)	-0.283*** (SE 0.082)	Before rotation control: negative pattern for Scola
RQ2 — same attacker selected ~ success + rotation	+0.246*** (SE 0.081)	-0.142 (SE 0.098)	Positive allocation pattern for Prandi
RQ2 — effect of rotation itself	-0.380*** (SE 0.083)	-0.263*** (SE 0.100)	Negative rotation effect in both teams
RQ3a — error ~ critical phase	-0.358** (SE 0.145)	-0.046 (SE 0.144)	Fewer errors for Brescia; no significant change for Talmassons
RQ3b — high ball ~ critical phase	-0.190** (SE 0.078)	-0.112 (SE 0.094)	Lower high-ball use for Brescia, no significant change for Talmassons

Table 2. Fixed-effect coefficients (β , log-odds scale) from logistic mixed-effects models with crossed random intercepts for player and match, estimated separately for each team. The two RQ2 models use the same sample ($n = 8,182$) and differ only in the inclusion of rotation. Significance: * $p < .10$; ** $p < .05$; *** $p < .01$.

6. Discussion

6.1 Performance and allocation behaviour

Taken together, RQ1 and RQ2 show a difference between attacking performance and allocation behaviour. Brescia shows no statistically significant real hot-hand effect, but after controlling for rotation, a successful attacker is more likely to be selected again on the team's next attack. This allocation pattern is therefore consistent with a hot-hand belief, even though RQ1 does not provide evidence of an actual hot-hand effect.

This result is similar to the experimental findings of Köppen and Raab (2012), where participants allocated more balls to players presented as being on a streak. However, it differs from Raab et al. (2012), who interpreted the use of streak information as adaptive because their data also showed evidence of real performance streaks. In the present study, no such effect is detected.

Scouting data cannot explain why the setter behaves in this way. The pattern could reflect a deliberate tactical decision, confidence in a particular attacker, or a hot-hand belief. The analysis can identify the allocation pattern, but not the psychological process behind it.

6.2 The importance of rotation

One of the main methodological findings of the analysis is the importance of rotation. In observational volleyball data, previous success and the attacking options available on the next action are not independent because winning a rally can cause the team to rotate.

This explains why the first RQ2 model produced a misleading result. Before controlling for rotation, Talmassons appeared less likely to select a successful attacker again, while Brescia showed no significant effect. After including rotation, the Talmassons effect disappeared and a positive effect emerged for Brescia.

This also shows an important difference between experimental and observational data. In controlled experiments, researchers can keep the available attackers constant. In real matches, player positions and attacking options change according to the rules of the game. These structural factors therefore need to be considered before interpreting allocation decisions.

The detailed court-position information available in Data Volley made it possible to identify and control for this factor.

6.3 Behaviour under score pressure

RQ3 provides no evidence that attacking performance becomes worse during critical phases. Talmassons shows no significant change, while Brescia makes fewer attack errors and uses fewer high balls.

The interpretation of high-ball use is important. The data show that high balls are strongly associated with the quality of the preceding reception, so their use should not simply be considered a more cautious attacking strategy. The lower high-ball probability for Brescia is instead consistent with the team reaching the attack in better conditions during critical phases.

However, this mechanism is not directly demonstrated. Positive reception increases slightly during critical phases, from 56.2% to 57.9%, but the difference is not statistically significant. A more detailed analysis of reception, defense, and transition phases would be needed to understand what produces the lower high-ball rate.

These results can also be compared with Ittlinger et al. (2025), who found biases in players' estimates of their probability of winning in elite beach volleyball. Their study concerns players' perceptions, while the present analysis measures observed performance and decisions. Together, these studies show that competitive pressure can be studied from different behavioural perspectives.

6.4 Implications

Although this study focuses on volleyball, the distinction between performance and allocation decisions can also be relevant in other contexts. Recent success may influence decisions about who receives the next opportunity, even when previous success does not necessarily predict future performance.

The results also show the importance of understanding the structure of the data-generating process. In this study, rotation substantially changed the interpretation of RQ2, while the relationship between reception quality and high-ball use changed the interpretation of RQ3. Behavioural patterns in observational data should therefore be interpreted together with the rules and constraints of the context in which they are generated.

7. Limitations and future work

The study has some limitations that could be addressed in future work. First, rotation is included only as a binary variable indicating whether a rotation occurred. Using the attacker's actual court position could provide a more precise representation of the attacking options available to the setter.

Success is also defined only as a direct point (#). A robustness check could include positive attacks (+) to test whether the results depend on this definition. In addition, RQ1 considers only the immediately previous attack, while some previous studies define streaks using two or three consecutive attempts.

Analysing longer sequences would allow a more direct comparison with this literature.

The mixed-effects estimator was validated using simulated data, but the results could also be compared with an established implementation as an additional check.

For RQ3, only 27.3% of attacks occur during critical phases, and the models estimate an overall effect that may differ between players. A possible extension would therefore be to investigate player-specific responses to score pressure.

Finally, the relationship between build-up quality and high-ball use is only partially examined in the current analysis. Future work could model reception and dig quality directly to test whether they explain the relationship between critical phases and the attacking situation.

8. Conclusion

This study separated the real hot hand from the behaviour associated with believing in it, using observational data from a full season of professional volleyball.

For RQ1, neither team shows evidence of a real hot-hand effect after accounting for differences between players and matches. For RQ2, the initial results changed substantially after controlling for rotation. The apparent negative pattern for Talmassons disappeared, while Brescia showed a significant tendency to select a successful attacker again. This behaviour is consistent with a hot-hand allocation pattern, even though RQ1 provides no evidence that previous success increases the probability of succeeding again. RQ3 also shows no evidence of worse attacking performance during critical phases. Talmassons remains relatively stable, while Brescia makes fewer attack errors and uses fewer high balls. Since high-ball use is strongly related to the quality of the preceding actions, this result should not be interpreted simply as a change in tactical risk.

An important result of the project is therefore methodological. The interpretation of the data changed when player heterogeneity, rotation, and the meaning of high-ball use were considered more carefully. These factors show that patterns observed in real match data cannot always be interpreted directly as behavioural effects. Understanding how the data are generated, including the rules and technical structure of the sport, is necessary before drawing conclusions about the decisions behind them.

Overall, the study finds no evidence of a real hot hand in either team, but it identifies an allocation pattern in Brescia that is consistent with a belief in the hot hand. At the same time, it shows how observational sports data can be used to study human behaviour, as long as the structural factors behind the observed patterns are taken into account.

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AI declaration

During the preparation of this report, the author used Claude and Grammarly for grammar and spelling checks, paraphrasing, language refinement, and to support the understanding and verification of theoretical concepts discussed in the course materials. These tools were used to improve the report's clarity and readability and to confirm the author's understanding of specific topics. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the publication's content.